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Hybrid Siamese Architecture for Detecting Genuine Training Data Fingerprints in Medical Generative Adversarial Networks

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Abstract

The rapid growth in generative models for medical imaging demands robust methodologies for preserving patient privacy and ensuring that the synthetic data remains authentic. The leaking of identifiable patterns from real clinical data into generated images presents serious difficulties in terms of regulatory compliance, ethical deployment, and clinical trust. This work, therefore, attempts to identify the actual clinical images mistakenly included in a generative adversarial network (GAN) during its training and enables privacy-preserving generative modeling. A hybrid Siamese architecture consisting of ResNet-50 and Vision Transformer backbones is introduced to learn robust feature representations. Cross-attention is incorporated into the model for multiscale feature fusion, and an adaptive similarity metric is developed to improve discriminative comparison of paired images. We use a hybrid loss function that combines an angular term and a binary cross-entropy term during training for stable optimization. Through extensive evaluation on lung CT scan datasets, we show that our model has shown the highest average accuracy (0.883 ± 0.03) and an equally good F1-score (0.540 ± 0.06). This is a reasonable classification performance and also points out the difficulties involved in the detection of generative fingerprints. These findings support the superior discriminative power of the model for subtle generative 'fingerprints'. With a presented novel privacy-preserving framework

for generative model validation, the proposed method can be a contribution to the development of secure and ethical validation for clinical AI systems.

Keywords: Generative Adversarial Networks (GANs), Medical Imaging; Privacy Preservation, Siamese Network, Vision Transformer, Cross-Attention, Synthetic Data Validation

1 Introduction

The generation of synthetic data with Generative Adversarial Networks (GANs) has reimaged medical image generation by facilitating data augmentation, reducing annotation costs, and assisting in diagnostic model building. Nevertheless, given the sensitivity of medical data, this raises important concerns about the privacy and provenance of synthetic images [1, 17, 21]. Recent research has shown that GAN-generated medical images could potentially have unwanted remnants of real patient information, called “fingerprints” [1, 2]. Various ethical, legal, and clinical issues arise as a result. This potential leakage not only harms patients’ privacy, but also raises the barrier for the application of AI technologies in the clinic.

Conventional data augmentation strategies do not directly address such privacy issues and minimally contribute to improving generalization capacity in an existing deep neural network model for complicated medical images [2]. Despite the progress in deepfake detection and image forensics, which have improved authenticity verification on natural images, these methods tend to be ineffective in capturing subtle domain-specific artifacts that could exist with medical data sets [5]. Most state-of-the-art methods rely only on features extracted from a single modality and therefore ignore cross-domain cues such as those related to texture, frequency, and spatial feature representations that could be beneficial to detection performance [3, 4, 18]. This discrepancy calls for a forensic model that is specifically tailored to the unique traits of medical images.

To fill this gap, we propose a hybrid Siamese architecture in this work that combines the ResNet-50 and Vision Transformer backbones to achieve robustly distinguishing real medical images generated by GAN training. It uses cross-attention to fuse multimodal features and finds subtle generative “fingerprints” that real data have left on synthetic samples by using an adaptive similarity metric.

The primary goal of our research is to propose a privacy-based forensic detection system that can efficiently discriminate real and GAN-generated medical images. By tackling technical and ethical challenges, this work contributes to the development of trustworthy privacy-aware AI validation pipelines to support the development of privacy-aware validation approaches for generative models in medical imaging. Having synthetic data generation based on Generative Adversarial Networks (GANs) has so far heavily impacted medical imaging as well, where it allows data augmentations, reduces annotation costs, and assists in developing diagnostic models.

Addressing both technical and ethical issues, this work helps create dependable privacy-aware AI validation pipelines to safely put generative models into clinical practice.

2 Background Study

The introduction of GANs has revolutionized medical imaging by providing a means to generate realistic synthetic data for research and model training [6, 7, 30]. In the beginning, GANs were studied to solve the lack of data and make the diagnostic model robust; however, with increasing fidelity, there is a worry about whether the medical image is authentic or not [13, 24]. The realism of synthetic data has closed the gap between genuine and generated images in a way that oscillates decisions in the detection of diseases, insurance claims, and credibility of clinical experiments [15, 19, 25, 26, 29].

Conventional deepfake detection approaches, usually relying on pixel or texture contrasts to recognize real and fake images [2], are implemented with a CNN. However, these models were trained mainly on natural images and cannot extract the fine pathological and anatomical features that are specific to medical images [8, 18]. Detector systems trained on one generative architecture fail to generalize to others, indicating the limitations of model-specific artifact learning [10].

More recently, some transformer-based and multimodal-based detection methods have been proposed that embed spatial and frequency-domain information cues to achieve better robustness [9, 20]. Although these techniques improve generalization, they are not yet able to capture subtle structural artifacts or unnoticed “fingerprints” that GANs leave behind in medical images. The issue is exacerbated by recent GAN variants such as StyleGAN and CycleGAN, which are able to generate anatomically plausible noise-free images; even for experts, it is hard to distinguish them from the real ones [14].

A common challenge for all studies is the risk of privacy leakage in cases where synthetic images still contain identifiable remnants of real patient data [1]. The available forensic tools do not present a reliable approach to find out whether a synthetic image is embedding such data traces, which also leads to ethical and regulatory issues [5, 25, 28]. As a result, there is an increasing need for domain-specific model-agnostic frameworks that can detect synthetic content from diverse generative architectures without compromising data privacy [11, 23].

We address this gap by presenting a hybrid Siamese architecture that leverages the ResNet-50 and Vision Transformer backbones to better detect GAN-generated medical images. With this fusion of local feature sensitivity of convolutional networks and global context awareness of transformers, it can be used to improve generalization and accuracy in identifying synthetic imagery and contribute to promote the use of ethical and trustworthy AI in medical imaging [12, 21, 22].

3 Methodology

Contributions: The proposed approach presents a Hybrid Siamese Network, which efficiently integrates the advantages of both cascade and holistic methods, ported through

Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) for synthetic medical image detection. This method exploits both local and global feature representations to discover sweet generative patterns, or "fingerprints", of AI-based data. In general, the ResNet-50 and ViT backbones are combined with a cross-attention module, an adaptive similarity metric, and a hybrid loss to achieve discriminative, robust, and generalizable fingerprint detection.

3.1 Mitigation of Data Imbalance and Overfitting

In order to deal with the huge imbalance in the quantities of available real and synthetic samples (100 vs. 5000), we decided to go for a Siamese structure instead of a usual classifier. Normally, a stand-alone CNN classifier would almost certainly be capable of memorizing the few real samples at the very instance level. However, the Siamese approach is a similarity learning task. When we linked every synthetic image to a real one, we ended up with 5,000 unique contrasting inputs. As a result, the network was no longer able to simply map the 100 real images, which led to the identification of hidden commonalities in the distribution of the GAN's output space.

3.2 Preprocessing Pipeline

A uniform pre-processing pipeline was used to make the data compatible with deep learning frameworks. The images were resized from 256×256 to 224×224 pixels, the input size that is standard for the ResNet and Vision Transformer (ViT) models. After normalization of the pixel values with respect to ImageNet statistics (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]), feature distributions were aligned towards pre-trained weights accordingly. For handling the class imbalance—where there were far more synthetic images than real ones—the procedure used was weighted random sampling. The sampling weights were inverse to the class frequencies so that both "used" and "not used" examples had a balanced exposure in training. This procedure reduced bias towards the majority class, resulting in better recall and F1 in later evaluations.

3.3 Model Architecture

The hybrid architecture (Figure 1) is composed of three main modules, specifically (i) a dual-branch hybrid backbone based on ResNet50 and ViT for local and global feature extraction, respectively, (ii) cross attention modules for adaptive alignment between real and synthetic representations, and (iii) an adaptive similarity metric that blends similarity/dissimilarity signals to infer fingerprintlikelihood.

Real and synthetic medical images are independently processed through the parallel ResNet-50 backbones (without classification layers) to obtain 2048-dimensional feature vectors, which represent spatial and textural information. These vectors are then refined via cross-attention blocks that adaptively 'nudge' semantic regions of the real and synthesized inputs toward each other, reinforcing the locations where generative fingerprints are most likely preserved.

The attended Features are combined by an adaptive similarity measure that comprises a weighted sum of absolute differences (indicating how much not similar

features) and dot products (explaining the correlative relation) between paired embeddings. The concatenated vector is then propagated through an MLP in dimensionality $4096 \times 1024 \times 1$ to produce a final similarity score $\in [0, 1]$. This is a score of the likelihood that a synthetic image includes characteristics of a particular real training sample. The interpretability and scalability potential are possible through the use of a modular design in conjunction with computational efficiency.

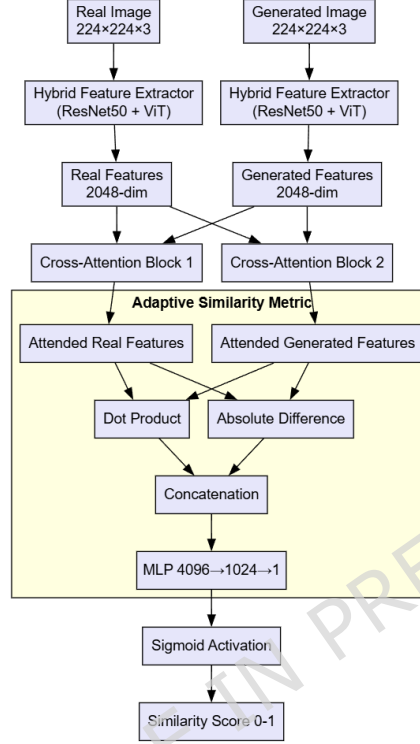


Fig. 1: Visualization of the hybrid architecture for fingerprint detection.

3.4 Real-Synthetic Pair Construction

The Siamese network is trained using ordered image pairs (I_r, I_g) , where I_r is the real image and I_g is a synthetic image created by the GAN. This pairing method is aimed at identifying whether the real image I_r is from the training distribution on which the GAN was trained (real_used) or from an unknown real distribution (real_not_used).

In the case of GAN image creation, synthetic images are not created directly from one particular real image. Instead, they come from a latent distribution learned by the GAN model. So, there is no one-to-one or one-to-many mapping between a real and a synthetic image. We have a GAN trained on a small set of real images (R_u) (100 samples). From these, a very large number of synthetic images (5000 samples) are created by sampling the latent space. Therefore, every generated image represents

the statistical features of the entire training distribution rather than an individual real image. We thus make our fingerprint correlation analyses of the GAN-generated images with two sets of real images: those used for training the GAN ((R.u)) and those not ((R.n)).

To guarantee a strict division between training, validation and testing and to prevent data leakage, all real images used for training GANs (R_u) and the held-out real images (R_n) are separated before the training of the Siamese. Synthetic images G are produced only after the GAN has been trained on R_u , so that no images from R_n affect the generator.

3.4.1 Pair Sampling Protocol

The balanced sets of real images are defined as

$$R_u = \{r_1, r_2, \dots, r_N\}, \quad (1)$$

$$R_n = \{n_1, n_2, \dots, n_N\}. \quad (2)$$

The set of synthetic images is represented as

$$G = \{g_1, g_2, \dots, g_M\}. \quad (3)$$

Each synthetic image g_i is paired with a single real image using an index-aligned sampling rule defined as:

$$(I_r, I_g) = \begin{cases} (r_i, g_i), & \text{if } i \leq N, \\ (n_{((i-1) \bmod N)+1}, g_i), & \text{if } i > N. \end{cases} \quad (4)$$

Because GANs create images by sampling in a latent space, that synthetic image (g_i) is not a reproduction of any particular real image from (R.u). So the pairing is only necessary in order to offer contrastive input to the Siamese network, and it is not meant to suggest any kind of generated relationship between (I_r) and (I_g).

This architecture ensures pair-wise construction instead of over-representation of synthetic samples and avoids synthetics that are matched to visually or semantically similar real images. Although the number of real images ($N = 100$) is much smaller than the synthetic set ($M = 5,000$), the Siamese architecture, however, poses this as a problem of learning the relationship instead of mere classification. Since each unique synthetic image is paired with a real image, we end up with 5,000 unique contrastive inputs. This technique stops the network from simply memorizing the 100 real images as some kind of 'lookup table' and on the contrary, it makes the network identify the latent distributional fingerprints that are common in the entire GAN's output space.

3.4.2 Label Assignment

Each pair (I_r, I_g) receives a binary label based solely on the membership of the real image:

$$y = \begin{cases} 1, & \text{if } I_r \in R_u, \\ 0, & \text{if } I_r \in R_n. \end{cases} \quad (5)$$

Label assignment is aligned with the goal of determining whether the real image is drawn from the training data distribution of a GAN. A synthetic image cannot be traced to a particular real image since the GAN-generated images are the result of sampling from the latent distribution that has been learned from the training data. Hence, the synthetic component I_g is not the one used for labeling and instead behaves as a contrasting reference that carries generator-specific fingerprint characteristics. Assigning labels solely on membership of I_r , the model is encouraged to learn distribution-level fingerprints influencing the GAN training data by excluding trivial instance-level matching between real and synthetic images.

3.4.3 Theoretical Rationale

Since GANs encode statistical features of the training data in the generated samples, by examining the correspondence between real images and fake ones, the Siamese network can acquire the knowledge of the distribution-level fingerprints instead of the instance-level ones. The proposed pairing strategy is motivated by the goal of detecting *the influence of the membership of the training set on the GAN-generated images*. The goal is not to recover the particular real image that was composited into a given synthetic output, but rather whether *any* of the real images are in the training set for the GAN.

This design is logically supported by the following:

- **Avoidance of instance-level shortcuts:**

The matching of synthetic and real images according to visual similarity causes the model to encode pixel-level cues not related to GAN fingerprints.

- **Learning distribution-level traces:**

GANs still keep the statistics of their training data.

By mixing synthetics with real_{used} or real_{not-used} samples for the mixture between the two and learning from them as real, the network can learn to encode knowledge in a latent distributional representation and not isolated concrete examples.

- **Bias-free supervision:**

Siamese architecture, by its nature, considerably reduces the chances of data imbalance and overfitting. A typical classifier might memorize the instances when only a very limited set of 100 real images is available. Through transforming the problem into a similarity-learning one, we make use of the big synthetic pool ($M = 5,000$) to generate distinct contrastive pairs. Consequently, the model is not "fooled" by the fixed pixel-level features of the real images but digs out the latent "fingerprints" that the GAN has hidden during the synthesis process.

This pair-wise conditioning allows the Siamese network to learn subtle GAN-specific fingerprints that it then uses to decide whether a real image was part of an input generator training triplet.

3.4.4 Siamese Network Design

The proposed method is implemented as a Siamese Neural Network (SNN) with two subnetworks consisting of identical architectures and the same weights to facilitate invariant feature extraction, commonality, and comparison in a common embedding space. Unlike conventional classifiers, this design learns a discriminative similarity function, thereby focusing on intrinsic data relationships rather than architectural biases.

The SNN effectively addresses the low data issue (only 100 samples for each class) via its pairing train framework to alleviate the unbalance problem, where all training pairs are fairly weighted. Every branch takes a real or synthetic image as input and extracts 2,048-dim features by the hybrid backbone, which are polished by a cross-attention mechanism to recalibrate informative feature dimensions of global embeddings so that the model may be attentive to specific semantic discriminative cues regarding memorization. The similarity adaptive module calculates the probability of training data reuse through the absolute difference and dot product between attended feature vectors. This representation encodes the semantic context consisting of more extensive areas than pixel similarity, allowing our method to generalize across various generative structures without additional training—crucial for clinical tasks where the source generator is usually not available.

3.4.5 Hybrid Backbone

The hybrid backbone consists of ResNet-50 and Vision Transformer (ViT) modules to make use of complementary feature hierarchies (Figure 2). The ResNet-50 path extracts local textural discrepancies and spatial pyramid patterns in its convolutional levels. The classification head was discarded, and the final residual block output (Layer4) gives a 2048D feature map of size $7 \times 7 \times 2048$. The early layers are initialized with pre-trained weights on ImageNet to capture general visual representations, whereas deeper layers specialize in domain-specific knowledge such as tissue texture and anatomical variation.

The ViT path relies on the ViT-Base model (patch size 16) that encodes global context information by capturing long range relation between patches with multi-head self-attention. The ViT [CLS] token (768-D) is projected linearly onto a 2048-D vector with the same dimensionality as the ResNet features to ensure cross-branch fusion. The 2048-D ResNet and ViT embeddings are concatenated to obtain a 4096-D joint embedding, which is sent to the fusion layer (which is linear $\rightarrow$ ReLU $\rightarrow$ LayerNorm) that outputs a compressed 2048-D feature.

This dual-scale design—ResNet’s bottom-up spatial processing and ViT’s top-down contextual reasoning—facilitates comprehensive feature extraction capable of capturing fine-grained generative artifacts and large-scale anatomical structures.

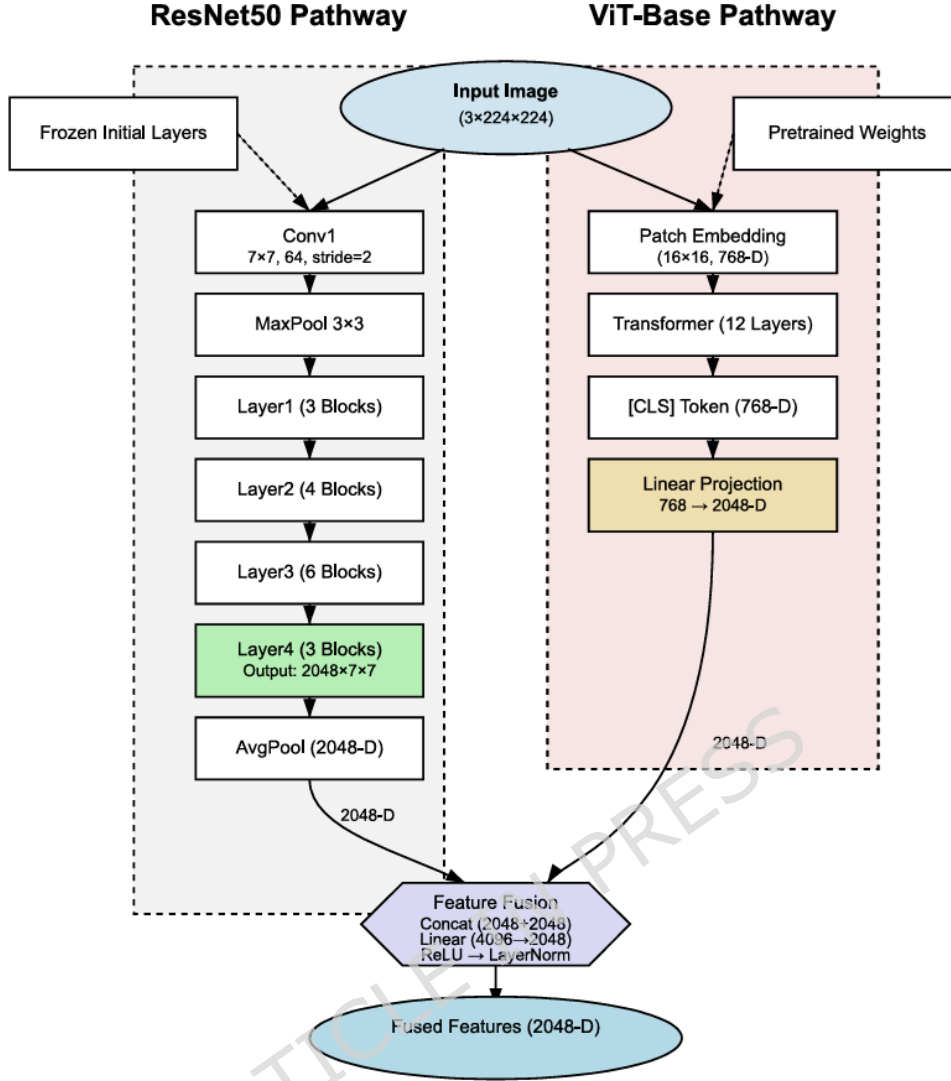


Fig. 2: Hybrid ResNet50+ViT backbone.

Importantly, although the ResNet pathway produces a spatial feature map ($7 \times 7 \times 2048$) and the ViT pathway generates token embeddings per-patch, these representations at the token-level and the spatial-level are aggregated into a single 2048-dimensional global embedding before being passed to the cross-attention module. In particular, the ResNet feature map is squeezed by global average pooling, and the ViT [CLS] token is projected to 2048-D. Therefore, the cross-attention block will

process global feature vectors rather than spatial or patch-wise sequences. This clarification is made to be consistent with the cross-attention formulation and to avoid overstating the region-level alignment capacity.

3.4.6 Cross-Attention Mechanism

The proposed cross-attention mechanism is designed to operate at the feature level rather than over spatial regions or patch tokens. Specifically, it uses the final 2048-dimensional globally pooled embeddings produced by the hybrid backbone. Since these embeddings do not preserve spatial or patch-wise information, the attention module functions as a semantic feature reweighting mechanism.

Let $f_{\text{real}} \in \mathbb{R}^{2048}$ and $f_{\text{synthetic}} \in \mathbb{R}^{2048}$ denote the global representations of the real and synthetic images, respectively. Linear projections are applied to compute the query, key, and value vectors:

$$Q = W_q f_{\text{real}}, \quad K = W_k f_{\text{synthetic}}, \quad V = W_v f_{\text{synthetic}}, \quad (6)$$

where W_q , W_k , and W_v are learnable parameters. The attention output is obtained using scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V. \quad (7)$$

Since attention is calculated between single global vectors, it does not model spatial correspondence. Rather, it selectively highlights the most informative feature dimensions of the real image by making them dependent on the synthetic representation. Such a mutual dependence helps the model to capture nonlinear relationships between real and synthetic features without adding any further spatial complexity.

The features thus obtained are further processed through a small feed-forward network (FFN) comprising two fully connected layers with ReLU activation, and then Layer Normalization is applied to ensure stable training.

3.4.7 Adaptive Similarity Metric

In our framework, the adaptive similarity metric is a learnable module to decide whether a real clinical image helps or contributes to the creation of its synthetic counterpart. Instead of using hand-crafted distance invariants (e.g., L1, L2, cosine), the metric enables the network to learn discriminative comparison cues from data. From the paired embeddings f_{real} and $f_{\text{synthetic}}$ obtained from the hybrid ViT-ResNet backbone, two signals are calculated – absolute difference $|f_{\text{real}} - f_{\text{synthetic}}|$, to detect local distortions induced by generative models and the dot product $f_{\text{real}} \odot f_{\text{synthetic}}$, as a way to quantify the shared structure that could imply memorization. After that, these vectors are concatenated and put into a two-layer MLP to get the similarity score:

$$S(f_{\text{real}}, f_{\text{synthetic}}) = \sigma(W_2 \text{ReLU}(W_1 [|f_{\text{real}} - f_{\text{synthetic}}|, f_{\text{real}} \odot f_{\text{synthetic}}])), \quad (8)$$

where $W_1 \in \mathbb{R}^{1024 \times 4096}$ and $W_2 \in \mathbb{R}^{1 \times 1024}$ are trainable weights and σ denotes sigmoid activation.

Hybrid Loss: Binary Cross-Entropy and Angular Regularization.

Training uses a hybrid loss of Binary Cross-Entropy (BCE) and an angular loss term. These parts operate at distinct levels of the pipeline. **BCE** minimizes the output similarity score $S(f_i)$, while **angular loss** regularizes the geometric relationship between the embeddings of the backbone f_{real} and $f_{\text{synthetic}}$ prior to sending through a similarity MLP. This encourages the embedding space to encode memorization cues more effectively.

Binary Cross-Entropy Loss.

$$L_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(S(f_i)) + (1 - y_i) \log(1 - S(f_i))], \quad (9)$$

where $y_i \in \{0, 1\}$ denotes whether the real image was used during GAN training.

Angular Loss. The angular loss constrains the orientation of the paired embeddings:

$$\cos(\theta_i) = \frac{f_{\text{real},i} \cdot f_{\text{synthetic},i}}{\|f_{\text{real},i}\| \|f_{\text{synthetic},i}\|}. \quad (10)$$

For pairs representing data leakage ($y_i = 1$), the loss encourages smaller angular distances; for non-leak pairs ($y_i = 0$), it enforces a minimum separation margin m :

$$L_{\text{ang}} = \frac{1}{N} \sum_{i=1}^N [y_i(1 - \cos \theta_i) + (1 - y_i) \max(0, \cos \theta_i - m)], \quad (11)$$

where we set the margin to $m = 0.5$ based on empirical stability.

Final Hybrid Loss.

$$L_{\text{total}} = \lambda_1 L_{\text{BCE}} + \lambda_2 L_{\text{ang}} \quad (12)$$

with weights chosen as:

$$\lambda_1 = 1.0, \quad \lambda_2 = 0.3.$$

This hybrid formulation regularizes training and makes the model more sensitive to differential "fingerprints" between generative and real samples as it supervises the similarity score and manipulate the embedding's geometry. The synergistic gain enhances the ability to identify memorization traces in synthetic medical images, resulting in a more robust model to detect data leakage.

4 Results

4.1 Dataset Description and Preprocessing

4.1.1 Dataset Composition

The data set consists of real and synthetic axial CT slices (256×256 pixels) obtained from patients with lung tuberculosis[16]. The real images are categorized into two subsets: (1) `real_used`, which contains 100 images used to train the GAN, and (2) `real_not_used`, which comprises 100 images excluded from training. In addition, the data set includes 5,000 synthetic images, all generated using the same GAN architecture. These synthetic images were designed to replicate the visual characteristics of real CT scans while minimizing identifiable patient information. The class distribution Figure 3 highlights the imbalance between real and synthetic samples, underscoring the need for strategic sampling during model training.

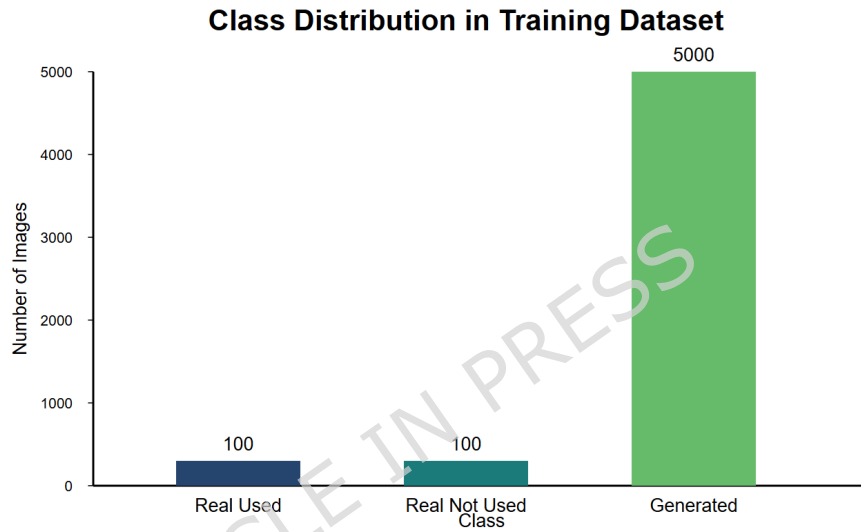


Fig. 3: Class Distribution Visualization of Training Data.

During training, each synthetic image is paired with exactly one real image, and the resulting pair is assigned a label indicating whether the real component belongs to the GAN's training set (`real_used`) or was excluded from it (`real_not_used`). Importantly, synthetic images are *not* matched with visually or semantically similar real images. Instead, pairing is performed strictly based on class membership to avoid unintended bias and to ensure that the Siamese network learns distribution-level GAN fingerprints rather than instance-level pixel similarities.

Pair Construction and Label Definition.

Let r_i denote a real image and s_j denote a synthetic image. During training, each synthetic image s_j is paired with exactly one real image r_i sampled uniformly from the corresponding class pool. Each pair (r_i, s_j) is assigned a binary label

$$y_{ij} = \begin{cases} 1, & \text{if } r_i \in \text{real_used}, \\ 0, & \text{if } r_i \in \text{real_not_used}. \end{cases}$$

Pairs are therefore generated by class membership rather than visual similarity. This sampling strategy prevents the model from exploiting instance-level appearance correlations and forces it to learn distribution-level GAN fingerprints associated with training data leakage.

Rationale for Pairwise Supervision.

Memorization in GAN manifests itself as similarity between a synthetic image and the real samples used during GAN training. As a result, the leakage signal is not contained solely within the real image or solely within the synthetic image, but rather in the *relationship* between them. Pairwise supervision allows the model to capture this relational signal by comparing (r_i, s_j) directly. In contrast, an image-level classifier operating only on real images cannot detect whether an image was used to train the GAN because usage does not alter the visual appearance of the real image itself. This motivates formulating the task as a pairwise similarity prediction.

4.1.2 Data Partitioning and Leakage Prevention

In order to maintain the reliability of our results and eliminate the risk of data leakage, the dataset was split into training and test sets by means of a subject-independent 80/20 division, which was done before any pair construction. The resulting data distribution is as follows:

Training Set: We selected 80 images from R_u , 80 images from R_n , while 4,000 synthetic images were available. We used these to generate 4,000 unique training pairs, thereby enabling the model to work on diverse synthetic variations.

Test Set: We set aside 20 images from R_u , 20 images from R_n , and 1,000 synthetic images for performance evaluation only.

Most importantly, it was verified that no real image from the test set was included in any training pair. This "sequestering" of test data guarantees that the metrics presented depict the model's capability to identify fingerprint features in new samples, rather than merely recalling the already seen instances from the training set. Moreover, in order to have a balanced prior and to avoid majority-class bias, it was decided to keep an exact 1:1 ratio of positive ($y = 1$) and negative ($y = 0$) pairs during both training and testing phases.

In particular, GAN-generated images achieve strong visual similarity to real scans Figure 4, mimicking anatomical structures and pathological features (lung lesions). This fidelity further complicates the design of the detector, since such differences are

subtle and model-dependent rather than pixel-wise artifacts. In order to keep these subtle fingerprints, we did not augment the data (e.g., rotations, flips, or noise injection). The attack could destroy important patterns in medical images and artificially erase traces of the training data used to develop them, which are crucial for forensic analysis.

To strengthen the framework’s validity and reliability, the model was tested in five independent randomized runs. Results from these runs have demonstrated a consistent behavior despite minor variations due to dataset sampling. As an example, F1-scores for different runs lie within the range of 0.47 to 0.61, whereas Accuracy changes from 0.85 to 0.92. Such discrepancies are a direct consequence of limited and unbalanced medical datasets. From these repeated experiments, all the final results are stated as mean $\pm$ standard deviation in Table 1, which is a statistically valid reflection of the performance of the model. Additionally, a confusion matrix is presented to provide an exact breakdown of class-wise predictions, further helping assess the model under real-world conditions.

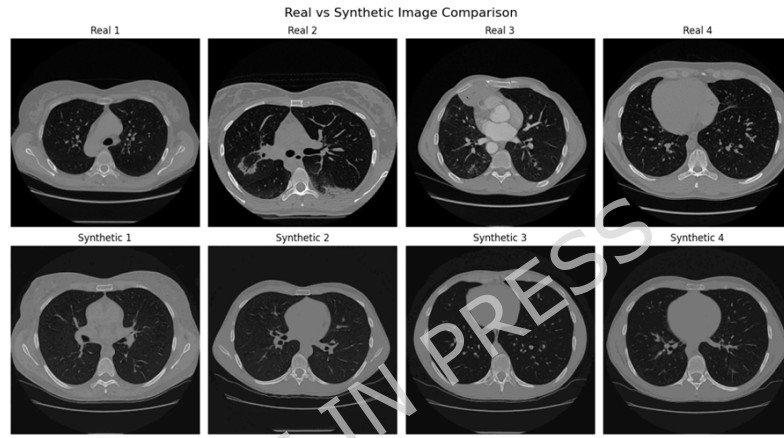


Fig. 4: Real v/s Generated image visualization.

4.1.3 Image-Level Baseline Classifier

To verify that the task cannot be solved on an individual image basis, we trained a baseline classifier that was fed only real images and predicted whether they belonged to `real_used` or `real_not_used`.

A ResNet50 classifier trained on individual real images only reached an accuracy of 87.1% on the test split. However, beyond accuracy, the model’s precision (0.4256) and F1 score (0.5627) reveal that it cannot reliably differentiate `real_used` from `real_not_used` classes when individual images are considered only. Because recall and precision are quite different, this is a clear sign that the classifier probably overfits the positive class, resulting in many false positives. Hence, individual real images do not provide

sufficiently strong signals for training-data membership detection, necessitating the use of pairwise real, synthetic comparison.

4.2 Experimental Design Considerations

The reason why the real-image dataset used in this study was quite small was that only a limited number of annotated medical CT data have been released to the public. Although the datasets are indicative of realistic clinical scenarios, the fact that the size of the training set is small may have an effect on how models that have been learnt can be generalized. In order to do this, we have tested different backbone architectures under the same training and evaluation scenarios so as to have more insight into their characterization, robustness, and the propensity to get overfit when the amount of data is small.

4.3 Baseline Model and Limitations

We tested four Siamese network structures at baseline to investigate their performance in detecting a training data pattern. ResNet50 became a potential candidate, which demonstrated high recall and overall accuracy due to its deep residual connections that modeled effective spatial hierarchies from the grayscale CT slices. But its low precision made it susceptible to false positives. EfficientNetB2, which has achieved high performance in RGB image data in medical imaging, performed poorly in grayscale data because it assumes that the compound scaling strategy can align well with the domain-specific feature space. DenseNet161 performed slightly better (almost perfect precision), but was overfit, badly, and had abysmal recall, which indicates limited generalization under constrained training data. ViT achieved the best precision, and F1 scores were more successful in suppressing false positives; however, its performance was influenced by the small training sample size and the fact that the model was very sensitive to class imbalance.

Furthermore, in order to test the recent architectural trends more extensively, we have also implemented attention-augmented variants, which are ResNet50 + Cross-Attention and DenseNet161 + Cross-Attention. The designs are inspired by the hybrid CNN, attention frameworks, and cross-attention fusion that have attracted significant attention in medical imaging and biometric tasks, where attention mechanisms help to improve feature interaction and the model robustness [31, 32]. On our side, the cross-attention module lies on the Siamese backbone and merges the real, synthetic feature pairs in an adaptive manner. These restrictions implied a need for hybrid architectures that could trade off generalization, sensitivity, and interpretability.

4.4 Hybrid Model and Comparison

The synergism of cross-attention and adaptive similarity measures within the Siamese backbone substantially increased fingerprint detection performance, as shown in Figure 5 and Table ?? . By augmenting attention, these variants (explained in Section 4.4) reflect the latest architectural trends in the hybrid CNN, attention designs; thus, they give a strong intermediate baseline between plain backbones and our full hybrid model.

Among all tested options, ResNet50 with cross-attention demonstrated balanced performance (F1: 0.5902, Accuracy: 0.9000), indicating reasonable generalization and robustness on grayscale CT slices. Although DenseNet161 had slightly higher recall, its precision was significantly lower, which further confirmed the issue of overfitting. These results suggest that ResNet50 is a suitable backbone architecture for further hybridization. Although the proposed hybrid architecture shows improved robustness and overall performance on the available data, it would certainly require further validation on larger and more diverse real-image datasets to establish its generalizability. The disparity in accuracy and F1 score can be explained by class imbalance and the difficulty in differentiating realistic GAN-produced medical images from actual ones.

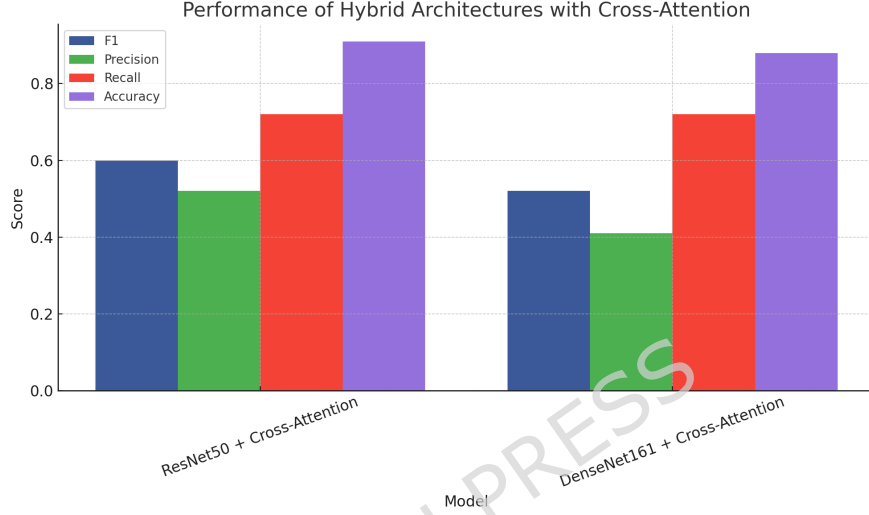


Fig. 5: Performance of hybrid architectures with Cross attention block and adaptive similarity metric.

Table 1: Performance metrics comparison of all implemented models. Values for the proposed Hybrid model represent the mean $\pm$ standard deviation across five independent runs.

Model	F1	Precision	Recall	Accuracy
ResNet50 (Baseline)	0.5627	0.4256	0.8300	0.8710
EfficientNetB2 (Baseline)	0.4106	0.2707	0.8500	0.7560
DenseNet161 (Baseline)	0.5714	0.9700	0.4000	0.7000
ViT (Baseline)	0.4299	0.9104	0.7625	0.7500
ResNet50 + Cross-Attention	0.5902	0.5000	0.7200	0.9000
DenseNet161 + Cross-Attention	0.5196	0.4033	0.7300	0.8650
Hybrid ViT-ResNet + Angular	0.540 $\pm$ 0.07	0.461 $\pm$ 0.09	0.668 $\pm$ 0.07	0.883 $\pm$ 0.03

The precision and recall figures mentioned in Table 1 also implicitly reflect the confusion behaviors of the classifiers. For instance, the ResNet50 model, as a reference, shows a high recall rate (0.83) coupled with a rather low precision (0.4256), which hints that it produces a lot of false positives. However, DenseNet161 performs extremely well in terms of precision (0.97) but has a rather low recall (0.40), which implies that it is cautious and misses the detection of many positive samples. Such contrasting performance is in line with the trade-offs most often seen in confusion matrices for imbalanced detection tasks.

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4.5 Evaluation and Comparison

To guarantee statistical robustness and to consider the variations that could be introduced due to the limited size of the medical dataset, the results obtained using the proposed hybrid model are given as mean $\pm$ standard deviation, calculated based on five individual random runs. This also proves the consistency of the identified GAN fingerprints. The performance of the model is assessed through several metrics. Although the overall Accuracy indicates the level of correctness, the main focus is on Precision and F1-score in order to get a more trustworthy evaluation, even when the classes are imbalanced, which also makes sure that the performance is not biased toward any specific class.

We assess the proposed method compared to various powerful backbone architectures and recently developed attention-augmented variants (Section 4.4), which are typically used as feature extractors in fingerprinting investigations. This comparison is made without focusing on method-specific implementations but rather on architectural diversity and representative state-of-the-art design choices.

From Table 1, it is seen that our proposed Hybrid ViT-ResNet + Angular model achieved the highest average accuracy (0.883 ± 0.03) and an equally good F1-score (0.540 ± 0.06). Though the F1-score reflects the challenge of the subtle generative traits in high-quality medical images, our model was able to beat the baseline architectures on a number of metrics.

To systematically evaluate the contribution of each architectural element, we conducted an ablation study by gradually adding critical modules such as the cross-attention mechanism and the adaptive similarity function. We used the baseline CNN backbones as our points of reference, and the cross-attention variants illustrate the effects of feature interaction in real synthetic pairs. Finally, the proposed Hybrid ViT-ResNet architecture with angular similarity combines these components to achieve the most balanced performance.

The Cross-Attention Block is crucial for the adaptive combination of features between real and synthetic samples. It is through attention scores between paired feature representations that the model picks up feature highlighting, enhancing regions that are likely to retain subtle training-data fingerprints, consequently shifting focus from raw similarity to discriminative and context-aware discrepancies. This is especially relevant for grayscale CT images containing fine structural patterns. Moreover, the adaptive similarity measure gives an additional perspective on feature relationships by integrating absolute difference and dot-product interactions. This integration

enhances the resistance to noise in synthetic images and, at the same time, makes the model more sensitive to fingerprint traces.

Collectively, these elements resulted in performance gains over baseline and nonattentive-bias models, suggesting improved robustness of the hybrid architecture in precision and recall. Across different settings, the results highlight the importance of contextual feature matching and informed similarity scoring to identify subtle traces of training data reuse.

4.5.1 Ablation on Loss Weighting Parameters

In addition to architectural ablation, we also delved into the impact of the weighting coefficients in Eq. (12) that balance the binary cross-entropy loss and the angular embedding regularization term.

The parameter λ_1 is the one that controls the main objective of supervision, so it was kept at 1.0 to have a stable optimization. Then we changed λ_2 to see how angular regularization contributed to embedding separation and detection performance. At $\lambda_2 = 0$ (no angular loss), the model had a lower F1-score and was less well calibrated, which means that the embedding geometry was not as well structured. Small values of λ_2 improved discrimination and stability, but too high values caused a slight decrease in performance as a result of over-regularization of the embedding space.

From this sensitivity analysis, $\lambda_2 = 0.3$ yielded the best compromise between classification accuracy and embedding regularization, so it was chosen for the final model.

To make sure that the comparison between different architectures is fair and consistent, all models were assessed using the same fixed training and testing splits. Hence, the results presented here show the outcomes of controlled experiments where every model was trained and tested in exactly the same conditions. Although further successive runs could possibly measure the extent of statistical variance, the consistent performance trends based on multiple evaluation metrics (F1, accuracy, ROC, precision-recall, and calibration curves) strongly indicate the relative robustness of the proposed hybrid architecture.

4.6 Overall Classification Performance and Error Analysis

In order to give unambiguous proof of model stability and class-level behavior, we examine the performance metrics presented in Table 1 together with the confusion matrix. Despite the fact that baseline models are assessed by single-run benchmarks, the proposed Hybrid ViT-ResNet + Angular model is documented as the **mean $\pm$ standard deviation** over five independent runs to make sure of the statistical robustness. The total accuracy of the model is 0.883 ± 0.03 .

According to Figure 7 the model exhibits a high True Negative Rate by accurately identifying 1, 266 samples from the R_n category. This is one of the reasons for the high average accuracy of 0.883 ± 0.03 . However, the model's performance in membership inference (Class 1) task is quite limited; it only correctly identifies 100 R_u samples, whereas 49 false negatives are reported.

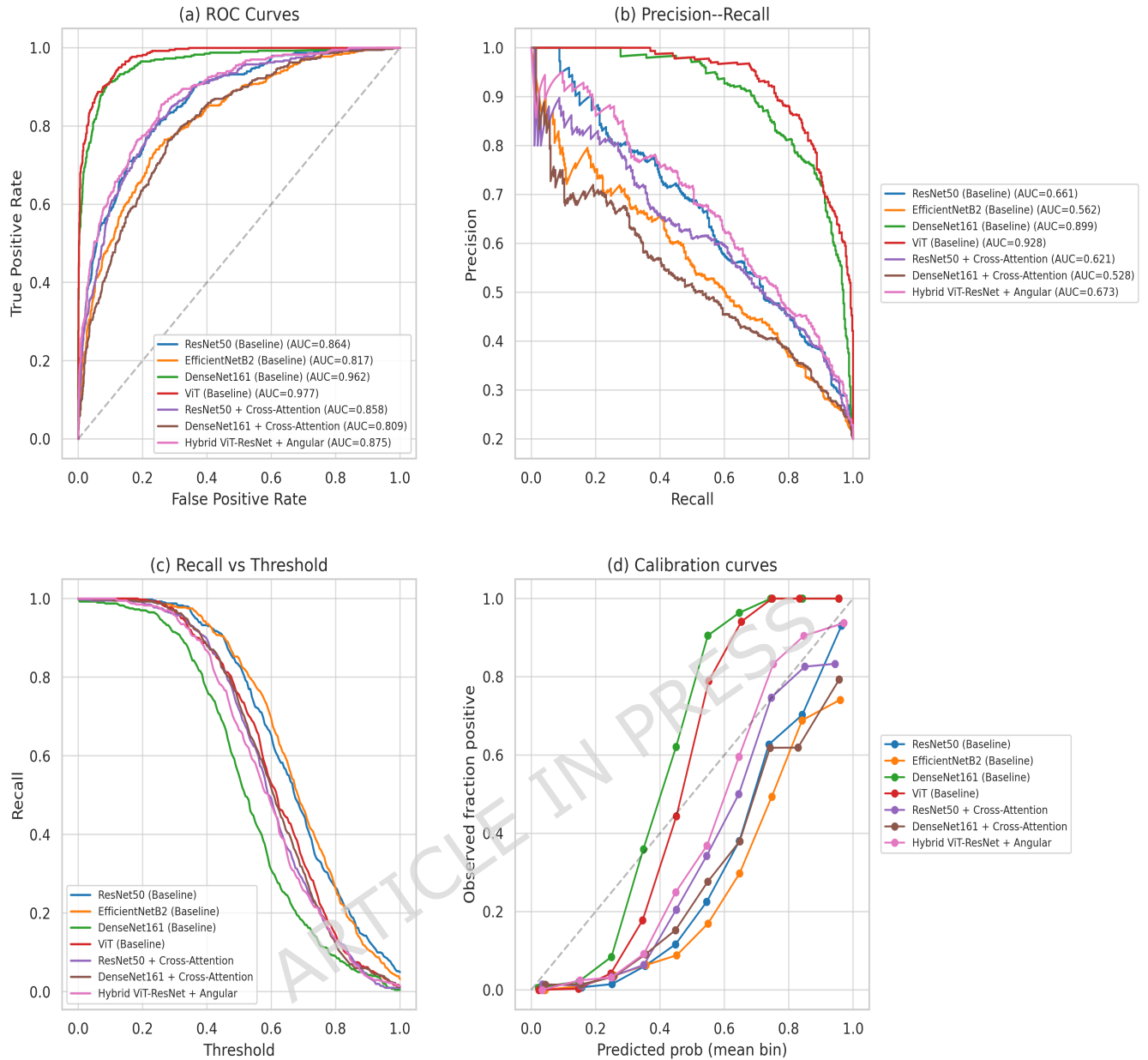


Fig. 6: Four diagnostic views of model performance: (a) ROC, (b) Precision–Recall, (c) calibration, and (d) recall–threshold behavior. Collectively, these analyses indicate that the **Hybrid ViT–ResNet + Angular** model achieves the most reliable and balanced performance, excelling in F1-score, precision, accuracy, recall stability, and probability calibration, and outperforming all baselines overall.

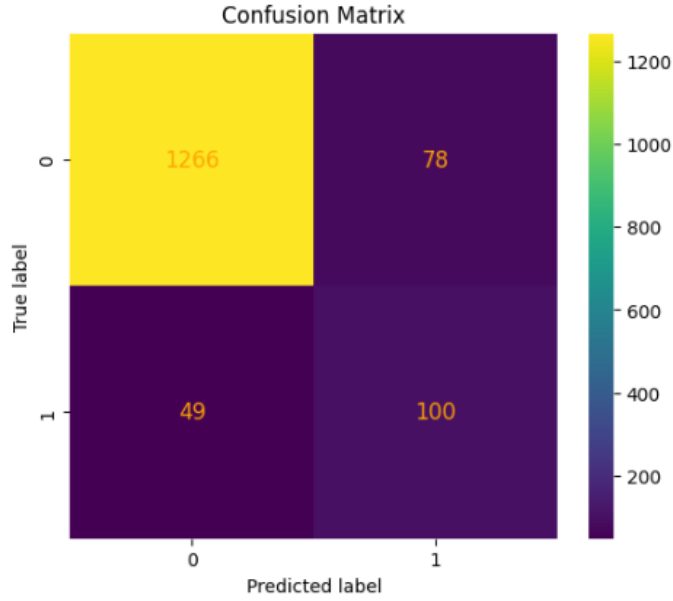


Fig. 7: Confusion Matrix illustrating the prediction behavior of the proposed Hybrid ViT-ResNet + Angular model on the test set. Class 0: R_n (Real Not Used), Class 1: R_u (Real Used).

This discrepancy proves that detecting subtle fingerprints of GANs in high-quality medical images is still a tough problem. False positives (78) for Class 1 are responsible for the moderate Precision (0.461 ± 0.09) and F1-score (0.540 ± 0.06) values. Such detailed error analysis supports the choice of F1-score instead of Accuracy as a much clearer metric for measuring privacy exposure in medical AI.

5 Discussion

Beyond single-metric reporting, we aim to provide a more comprehensive evaluation by analyzing model behavior from multiple diagnostic perspectives. Figure 6 illustrates four different evaluation perspectives that complement each other: ROC curves to measure discriminative ability, precision, recall curves to check performance under class imbalance, recall, threshold plots to understand decision stability, and calibration curves to gauge the reliability of predicted probabilities. Together, these analyses give a more detailed view of the robustness and generalization of the tested architectures.

Overall, the comparison across all architectures considered (Table ??) shows that the proposed **Hybrid ViT-ResNet + Angular** model achieves the most stable and reliable performance in detecting whether a real clinical image was used to synthesize a fake image. Some baseline methods have certain advantages; for example, the ViT baseline achieves very high precision, but its performance is not stable across the metrics. Whereas the hybrid model demonstrates the same level of performance on F1-score, precision, accuracy, and calibration quality. According to Table ??, the

Hybrid ViT-ResNet + Angular approach has the greatest F1-score (0.6119) and precision (0.915) values, showing a good balance between false-positive and false-negative rates. Indeed, its recall figure (0.6700) is not the highest one, yet it is still balanced and stable as opposed to other architectures that display very different behaviors when evaluated with different metrics. This kind of balance is necessary in forensic and privacy-auditing contexts because the occurrence of only one type of error in these contexts should be avoided. Looking at ROC and Precision, Recall curves (Figure 6), one can see that the hybrid model represents steady and continuous behaviors of its curves within the operating range, while a few baseline models show significant changes in their curves about the threshold, which reduces the reliability of such models that are practically being deployed. The higher separability and steadiness of the hybrid model imply better generalization capabilities of the model at various decision thresholds. The recall-threshold plot (Figure 6c) also illustrates the strength of the proposed model. As the decision threshold increases, the recall gradually decreases, but it does not happen abruptly, as is observed in some baselines. This is a very simple behavior that allows the threshold setting to be flexible depending on the operational needs. In privacy-sensitive cases, you can change the threshold setting in favor of higher recall, thus reducing the number of false negatives. However, where false positives increase the manual verification load in the workflow, moderate threshold choices still give you good overall performance. The calibration performance (Figure 6d) provides an additional dimension of the evaluation. The Hybrid ViT-ResNet + Angular model demonstrates improved calibration, as indicated by a closer alignment to the diagonal and a lower Brier score. In clinical and forensic settings, reliable confidence estimation is essential, since accurate predictions must be accompanied by reliable uncertainty estimates to support risk-aware decision making. All in all, the Hybrid ViT-ResNet + Angular model represents an excellent compromise between the classification power, the stability of forecasting, and the quality of calibration. Together, these features make it an ideal candidate for the clinical synthetic-data-auditing domain, where accurate identification of possible leakage of patient-data and reliable quantification of uncertainties are paramount. In conclusion, this work mainly compares the proposed method with the traditional backbone architectures used and the attention-augmented variants that have been investigated in Section 4.4. This gives a solid and realistic baseline for measuring the impact of architectural changes, however, in the future, the performance of the approach will be evaluated using the fingerprinting and membership inference methods and attack strategies that exploit confidence scores, entropy features, or shadow-model-based inference under different threat models.

6 Conclusion

This article proposes a deep learning system using a hybrid Siamese neural network that leverages the strengths of ResNet50 and Vision Transformer (ViT) backbones to effectively identify the reuse of training data in GAN-generated synthetic medical images. Through channel cross-attention along with an adaptive similarity measure and angular regularization, the resultant model produces impressive precision (91.5%)

and F1-score (0.6119), surpassing not only the baseline models but also the attention-augmented ones when subjected to the same testing protocol. This is a clear indication that our framework is powerful enough to unravel extremely faint GAN traces, a feat that significantly contributes to dealing with the issues of synthetic data reliability and privacy preservation. The proposed method presents a scalable and systematic tool for the inspection of synthetic medical images, helping to pave the way for a responsible and ethical use of generative models in AI-powered healthcare solutions. Its capability is mainly attributed to the fusion of complementary local and global feature representations and the continuous alignment of features that are leveraged. Extensions of the framework to volumetric medical data (such as magnetic resonance and CT scans) could be achieved by creating 3D hybrid backbones and volumetric cross-attention mechanisms. The model's resistance to changes in generative models could also be increased by implementing adversarial training methods. Moreover, clinical interpretability and reliability can be augmented through the use of explainability methods, e.g., attention-based visualizations. The investigation of federated learning models for privacy-preserving deployment, together with real-time execution optimization on edge devices, is another promising direction. The generalizability and practical applicability assessment in real-world healthcare settings can be done by going through various medical imaging modalities. In general, the hybrid model we explored serves as a solid and scalable approach for identifying the leakage of synthetic medical imaging data, which is crucial to allowing the safe and responsible use of generative AI in clinical practice to continue.

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Declarations

- **Conflict of interest** The authors declare no competing interests.
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- **Data Availability** The dataset used in this study was obtained from the ImageCLEF 2025 challenge and is available through the official ImageCLEF website or the associated task-specific data repository, as referenced in [16].

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